



Technical Service Bulletin: TSB210023

Released Date: 25-Feb-2021

Choosing Correct Exhaust Gas Recirculation (EGR) Temperature Sensor with Replacement of Main Engine Wiring Harness

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Products Affected

- X15 CM2350 X114B - Efficiency Series
- X15 CM2350 X116B - Performance Series

Description of Change

- The exhaust gas recirculation (EGR) temperature sensor and engine wiring harness designs of the products affected have changed.
- Because of differences between the existing EGR temperature sensor design and the new EGR temperature sensor design used on the products affected, there will be an incompatibility encountered when attempting to reconnect the existing EGR temperature sensor, Part Number 2872140, to the replacement engine wiring harness, Part Number 5484117.
- A new EGR temperature sensor, Part Number 5461596, will need to be ordered if installing replacement engine wiring harness, Part Number 5484117, and the engine is equipped with the existing EGR temperature sensor, Part Number 2872140.

Service Instructions

If the existing EGR temperature sensor requires replacement due to engine wiring harness compatibility, see Figure 1 below and Routing Instructions, to securely route and tie down new EGR temperature sensor lead cable.

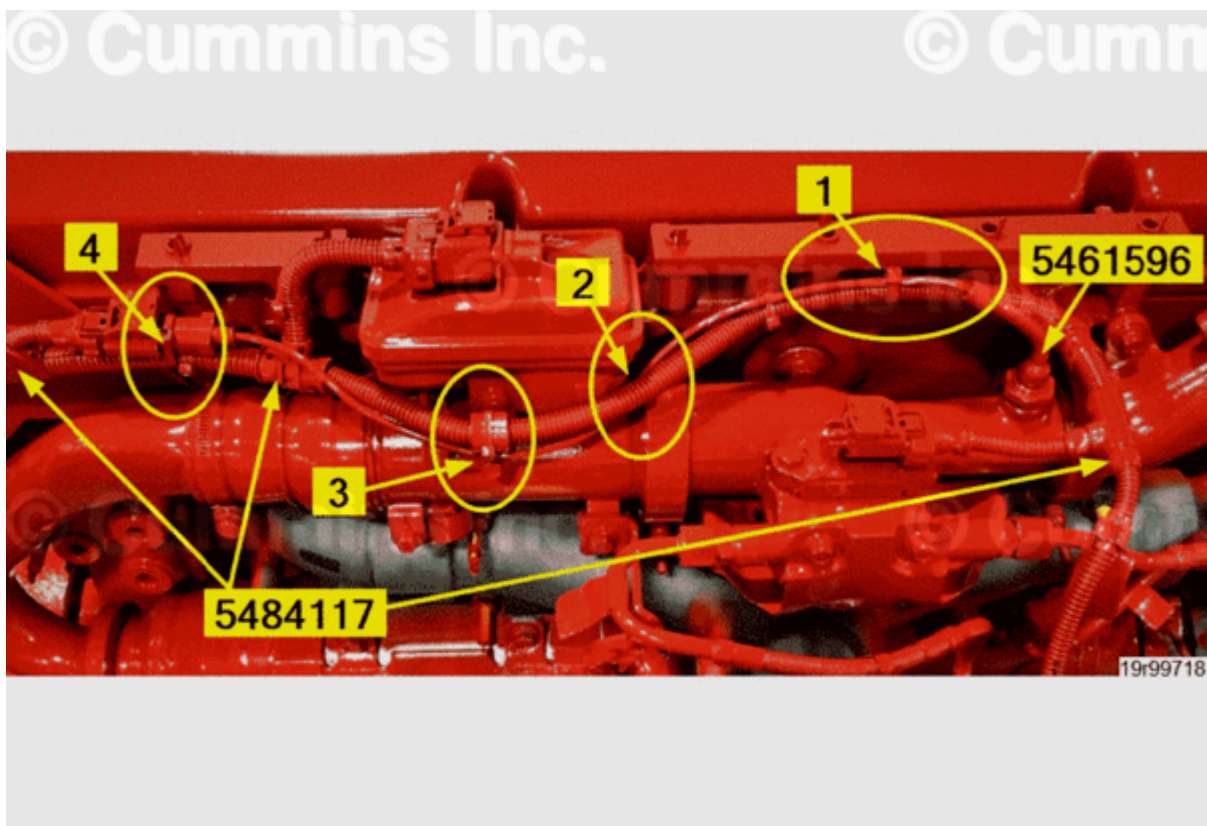


Figure 1, New Engine Wiring Harness, Part Number 5484117 and New EGR Temperature Sensor, Part Number 5461596.

Routing Instruction

1. Route cable on top of engine wiring harness and use one wire tie, Part Number 3062329, or equivalent.
2. Route cable behind engine wiring harness as shown.
3. Position cable underneath P-clip as shown, connect to engine wiring harness connector and secure secondary (red) latch.
4. Tie down EGR temperature sensor interconnect to harness exactly at location shown using one wire tie, Part Number 3062329, or equivalent.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
SENSOR, TEMPERATURE	2872140	No	Yes	5461596
HARNESS, ETR CNT MDL WRG	3692964	Yes	Yes	5484117
CLAMP,WIRE TIE	3062329	No	No	No

Part Compatibility

Visual differences between the two versions of EGR temperature sensors are as follows:



Figure 2, Existing EGR Temperature Sensor, Part Number 2872140.

- The existing production EGR Temperature Sensor, Part Number 2872140, as shown in Figure 2 above has a wire lead assembly made up of two individual white Teflon insulated wires encased within a loose fitting, woven sleeve. There is a short, grey silicone rubber wire seal found at the top of the sensor casing body.
- The lead cable on the existing sensor has a shorter wire lead assembly cable length. The lead cable will not reach its respective connector on the new engine wiring harness , Part Number 5484117.

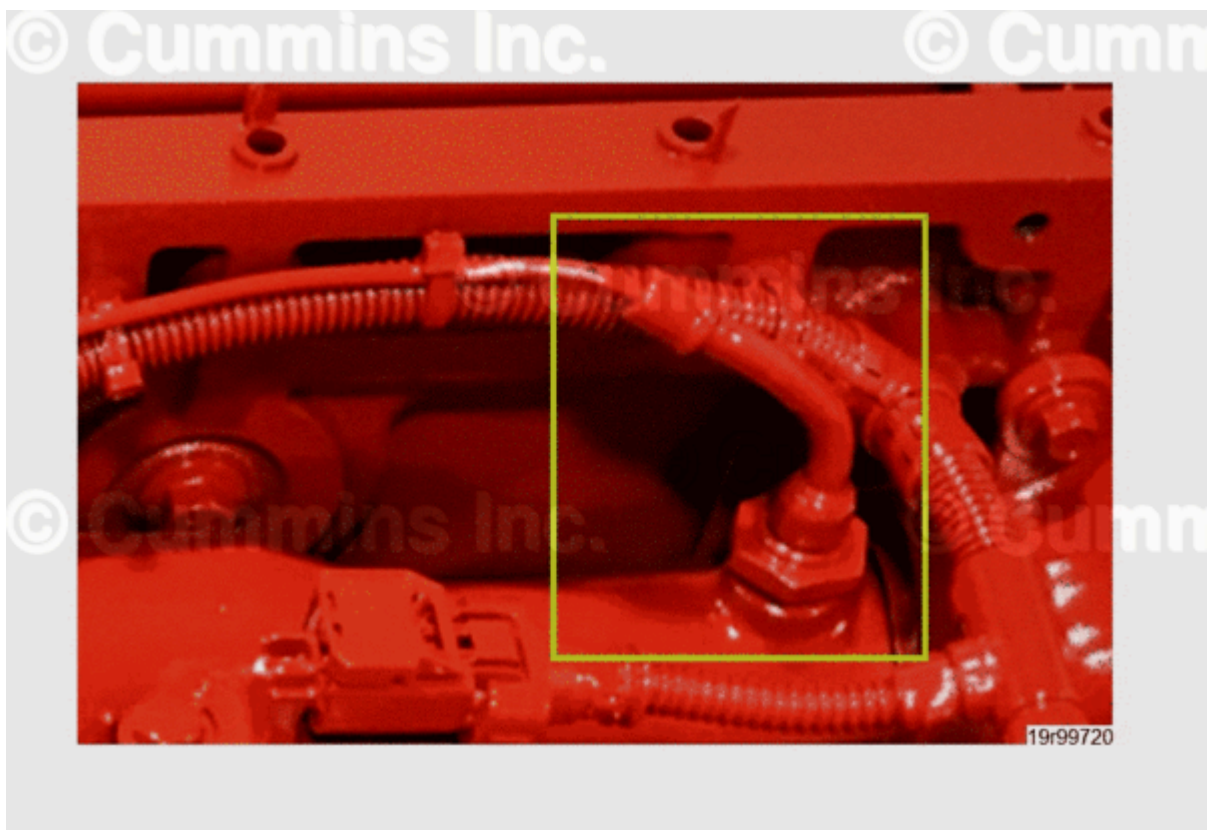


Figure 3, New Production EGR Temperature Sensor, Part Number 5461596.

- Figure 3 above shows the new production version of the EGR temperature sensor, Part Number 5461596. The sensor connector **must** be held in place with one wire tie as shown above in Figure 1.
- This new sensor uses a single jacketed, silicone rubber-insulated cable with a long grey, flexible silicone rubber strain relief/seal component.
- Existing and new EGR temperature sensors will have their respective part numbers stamped upon the stainless-steel casing below the hex flats.

Table 2, Production Information: EGR Temperature Sensor, Part Number 5461596

ESN First	Build Date ¹	Plant
80019410	24 October 2017 and after	Jamestown Engine Plant

¹Engine build date can be found on the engine dataplate.

Table 3, Production Information: Main Engine Harness, Part Number 5484117

ESN First	Build Date ¹	Plant
80117501	29 October 2018 and after	Jamestown Engine Plant

¹Engine build date can be found on the engine dataplate.

Publications Affected

Table 3, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	X15 CM2350 X114B	5411181	Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-019-043.html)	19
Service Manual	X15 CM2350 X114B	5411181	EGR Temperature Sensor	Refer to Procedure 019-378 (/qs3/pubsy s2/xml/en/pr ocedures/37 7/377-019-378.html)	19
Service Manual	X15 CM2350 X116B	5411186	Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-019-043.html)	19
Service Manual	X15 CM2350 X116B	5411186	EGR Temperature Sensor	Refer to Procedure 019-378 (/qs3/pubsy s2/xml/en/pr ocedures/45 3/453-019-378.html)	19

Document History

Date	Details
2021-2-3	Module Created

